

$$s \sim P_{\text{expert}}$$

$$\pi_{\theta}(s_t) \neq \pi_E(s_t)$$

 s_t^E
 ϵ_t
 t

$$s \sim P_{\pi_{\theta}}$$

 s_t^{π}

$$\min_{\theta} \mathbb{E}_{\mathbf{s} \sim P_{\text{expert}}} [\|\pi_{\theta}(\mathbf{s}) - \pi_{\text{expert}}(\mathbf{s})\|_2^2], \quad P_{\pi_{\theta}} \neq P_{\text{expert}}$$